MILITARY SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, ELECTRICAL, CLASS I

(COAXIAL, SERIES TWIN, TWTNC, BULKHEAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connector described herein shall consist of this specification sheet and the latest issue of MIL-C-3655.

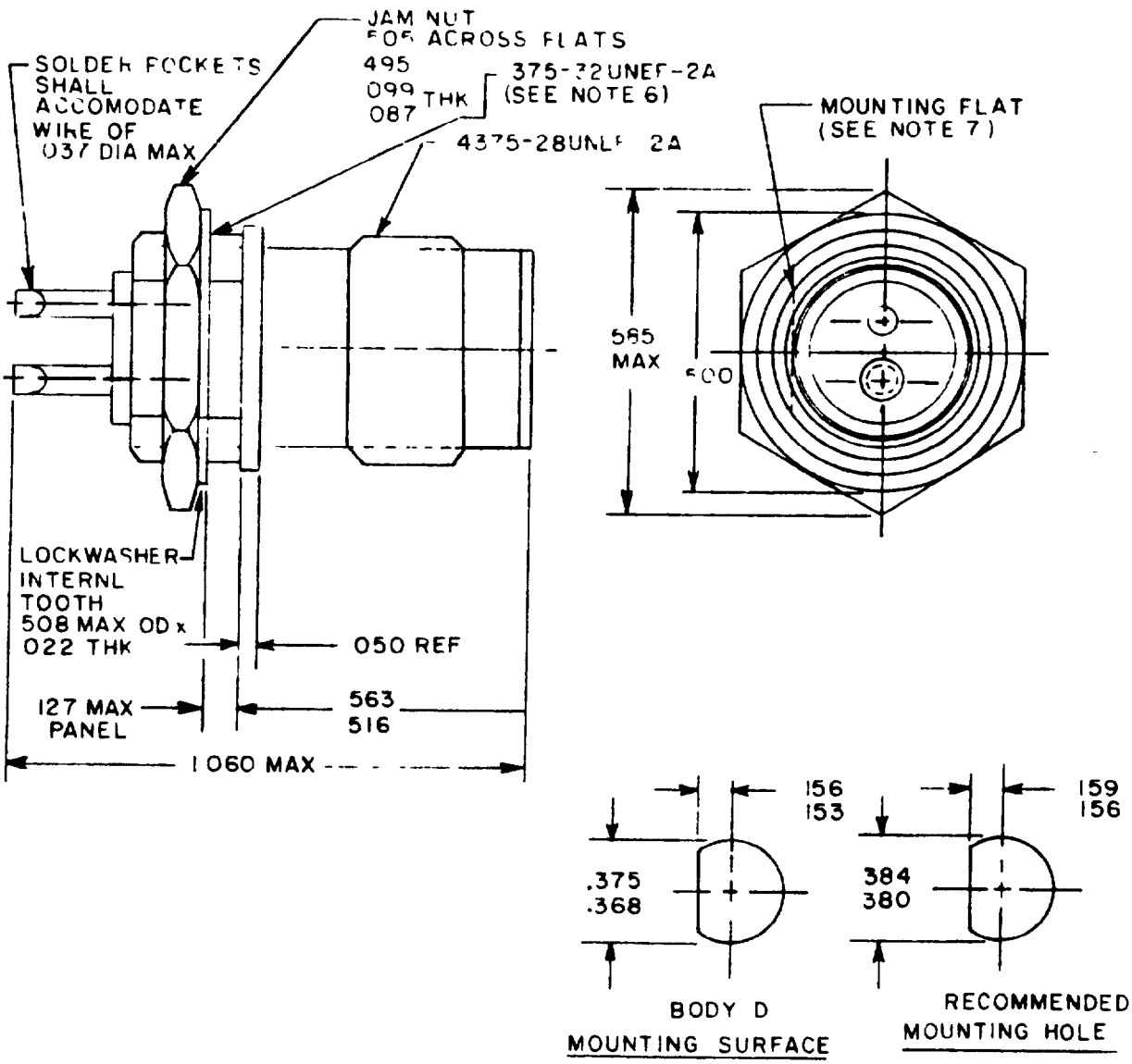


FIGURE 1. Type TWTNC, electrical bulkhead receptacle connector.

FSC 5935

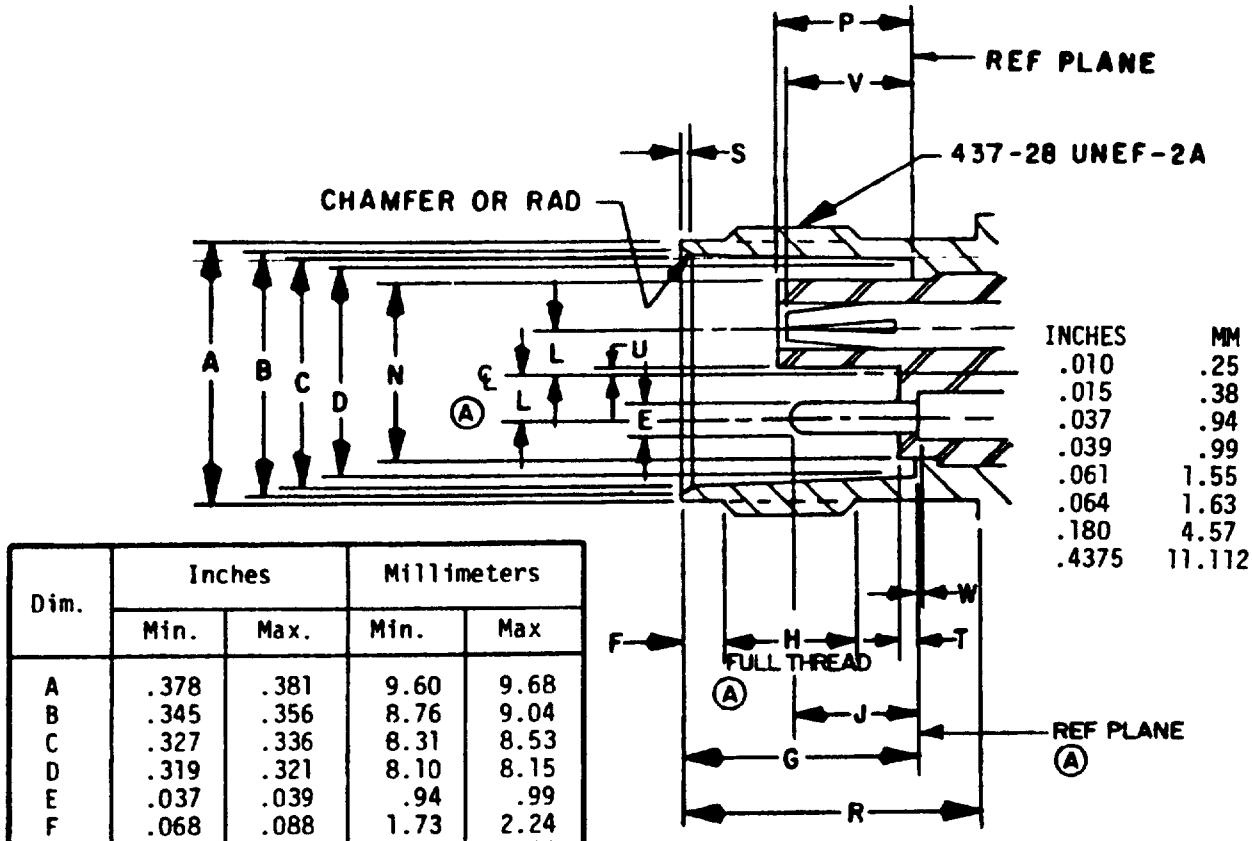
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INCHES	MM
.005	.13
.022	.56
.050	1.27
.087	2.21
.099	2.51
.127	3.23
.153	3.89
.156	3.96
159	4.04
.16	4.1
.368	9.35
375	9.52
380	9.65
.384	9.75
.4375	11.112
.495	12.57
.500	12.70
.505	12.83
.508	12.90
516	13.11
.563	14.30
.585	14.86
1.060	26.92

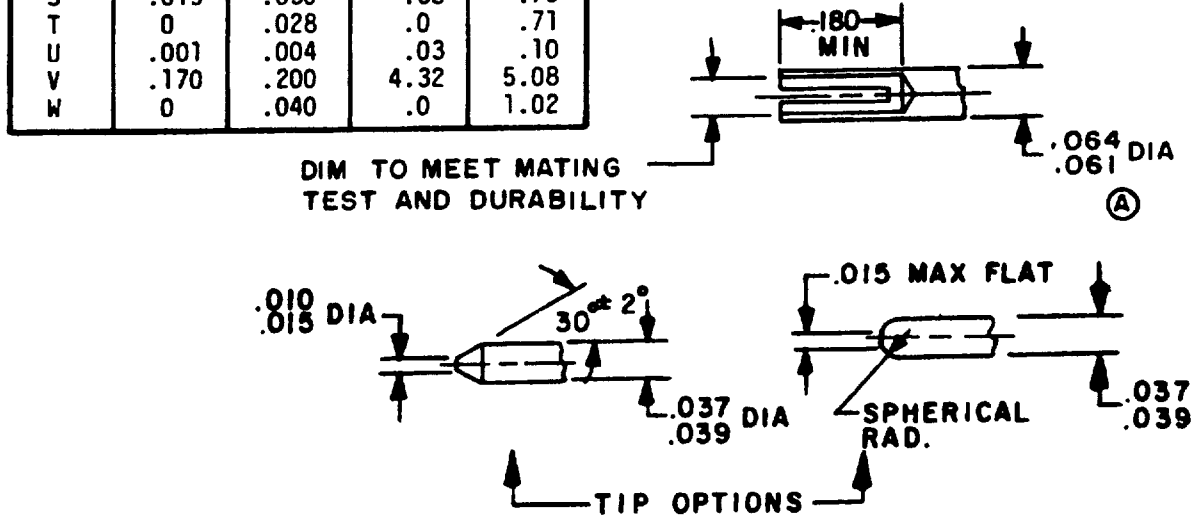
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3 Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Wrench flats are to accommodate standard wrench openings per H-28, app. 10.
5. All undimensioned pictorial representations are for reference purposes only.
6. Full threads to within .063 of shoulder.
7. Contacts, insulator and mounting flat shall be oriented within ±3° of position shown.

FIGURE 1. Type TWTNC, electrical bulkhead receptacle connector - Continued.

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Dim.	Inches		Millimeters	
	Min.	Max.	Min.	Max
A	.378	.381	9.60	9.68
B	.345	.356	8.76	9.04
C	.327	.336	8.31	8.53
D	.319	.321	8.10	8.15
E	.037	.039	.94	.99
F	.068	.088	1.73	2.24
G	.329	.333	8.36	8.46
H	.187	-	4.75	-
J	.171	.200	4.34	5.08
L	.062	.064	1.57	1.63
N	-	.262	-	6.65
P	.188	.206	4.78	5.23
R	.415	-	10.54	-
S	.015	.030	.38	.76
T	0	.028	.0	.71
U	.001	.004	.03	.10
V	.170	.200	4.32	5.08
W	0	.040	.0	1.02



CONTACT DETAILS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Type TWTNC, visual mating requirements.

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ENGINEERING DATA:

Nominal impedance. 78 ohms.

Frequency range. 0 to 500 MHz.

Voltage rating: 500 Vrms, maximum, at sea level; 125 Vrms, maximum at 70,000 feet.

Mating connector: M3655/14-0001.

REQUIREMENTS:

Design and construction: See figures 1 and 2.

Force to engage/disengage:
Torque: One inch-pound, maximum.
Longitudinal force: 3 pounds, maximum.

Coupling proof torque: Not applicable.

Mating characteristics (see figures 1 and 2 for dimensions):
Inner contact (sockets):
Oversize test pin: 0.042 inch diameter, minimum (nonclosed entry contacts only).
Test pin finish: 16 microinches.
Insertion depth. .093 inch, minimum.
Number of insertions: One.

Insertion force test.
Steel test pin diameter: .039 inch, minimum.
Test pin finish: 16 microinches.
Insertion depth: .093 inch, minimum.
Insertion force: 2 pounds, maximum.

Withdrawal force test:
Steel test pin diameter: .037 inch, maximum.
Test pin finish. 16 microinches.
Insertion depth: .093 inch, minimum.
Withdrawal force: 2 ounces, minimum.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance: 5×10^9 ohms, minimum.

Contact insulator captivation: Applicable.

Salt spray (corrosion): Applicable.

Connector durability: Applicable.

Contact resistance (milliohms, maximum):

	Initial	After environment
Inner conductor contacts:	3.5	4.7
Outer conductor contacts:	.4	Not applicable

Dielectric withstanding voltage: 1,500 Vrms, minimum, at sea level.

Vibration, high frequency: Applicable.

Shock (specified pulse): Applicable.

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Temperature cycling: Applicable.

Humidity: Applicable, but no measurement shall be made at high humidity. Insulation resistance shall be at least 2×10^8 ohms within 5 minutes after removal from the humidity chamber.

Cable retention force: Not applicable.

Coupling nut retention. Not applicable.

Marking: In accordance with MIL-C-3655.
Part number: M3655/23-0001.

Qualification: Required.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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MIL-C-3655/22A CONNECTOR, RECEPTACLE, ELECTRICAL, CLASS I

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Ft. Monmouth, N. J. 07703

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